Honeywell

DeepTrace

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Explainable AI-Powered Behaviour Intelligence for Enterprise Intrusion Detection

Design and build an AI/ML system that models normal access and connection behaviour for users and devices, detects intrusions or compromised-credential activity in near real-time, and classifies anomalies with an explainable risk score.

Project Details

Theme: Cybersecurity

Category: Software

Submitted By

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PROBLEM MAPPING

Proposed Solution

DeepTrace is an explainable AI-powered behaviour intelligence platform that learns normal user and device behaviour from sequential access events, detects cyber threats in near real-time, adapts to behavioural drift, and explains every prediction using SHAP-based risk attribution for enterprise SOC analysts.

Enterprise Challenge	DeepTrace Module
Sequential Behaviour	**Transformer Behaviour Encoder**
Class Imbalance	**Isolation Forest**
Concept Drift	**Behaviour Evolution Module**
Explainability	**SHAP Explainability**
Cold Start	**Behaviour Similarity Engine**
Near Real-Time Detection	**Threat Fusion Engine**

Sequence-Based Learning

Learns temporal behavioural patterns

Hybrid AI Detection

Transformer + Isolation Forest

Adaptive Concept Drift

Continuously updates behavioural baseline

SHAP Explainability

Transparent feature attribution

Cold-Start Profiling

Similarity-based user/device initialization

Enterprise SOC Dashboard

Interactive investigation workspace

SYSTEM ARCHITECTURE

System Architecture & Implementation

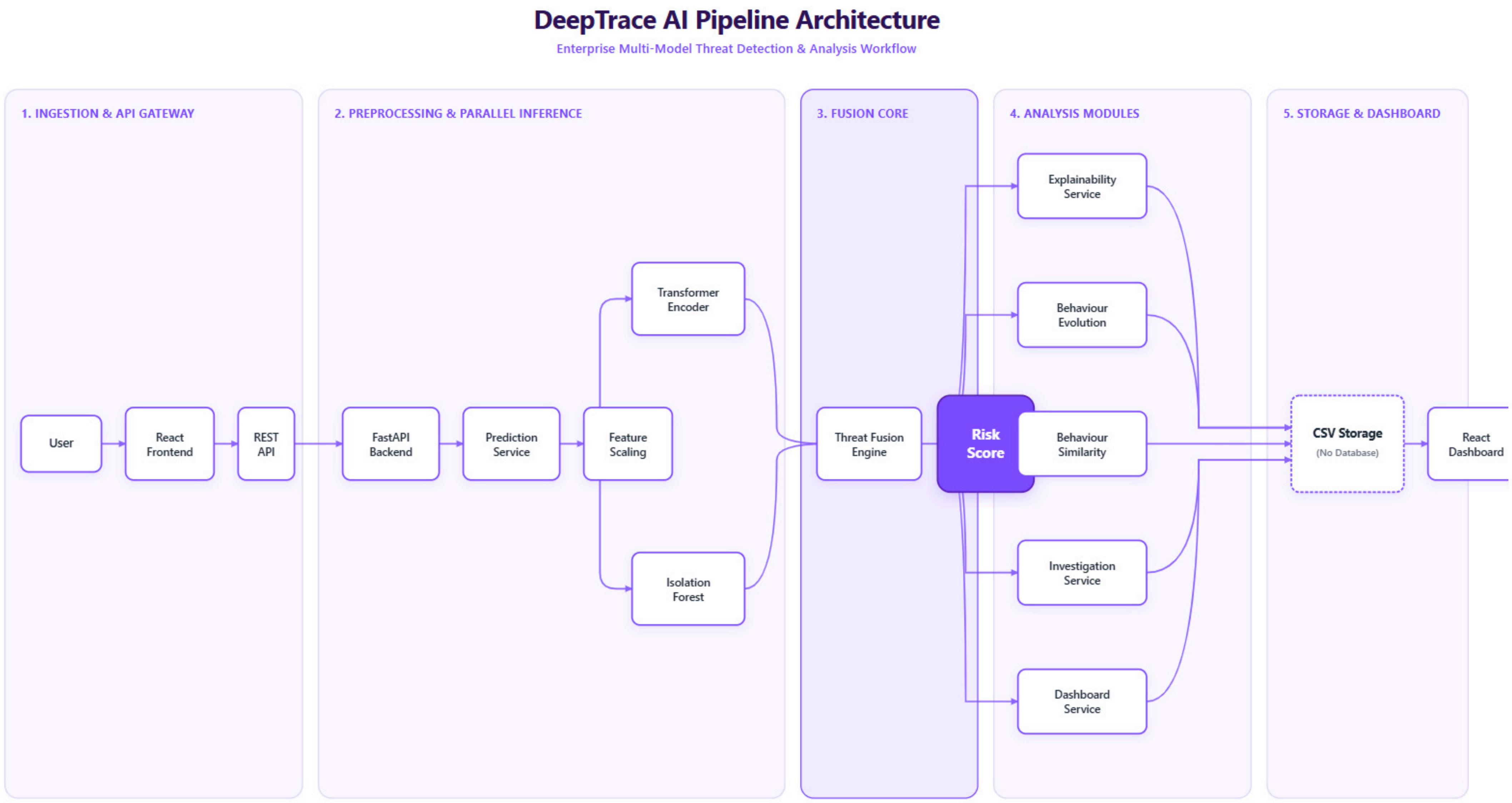


Figure: DeepTrace Enterprise AI Pipeline – End-to-End Behavioural Threat Detection Workflow

Technologies Used

Programming Languages
Python • TypeScript

Frameworks
React • FastAPI • Vite • Tailwind CSS

AI / ML
PyTorch Transformer • Isolation Forest • SHAP Explainability

Visualization
Plotly • Chart.js

Hardware
Standard CPU • Optional GPU Acceleration

Implementation Methodology

Behaviour Telemetry → REST API → Feature Preprocessing → Parallel AI Models → Threat Fusion → Behaviour Analysis → Dashboard

Behaviour telemetry is processed through a modular AI pipeline where Transformer and Isolation Forest perform parallel inference to generate explainable risk insights in near real time.

FEASIBILITY ANALYSIS

Feasibility & Viability Analysis

DeepTrace is a technically feasible and scalable solution built on open-source technologies. Its modular AI architecture supports enterprise deployment while enabling explainable, adaptive, and near real-time behavioural threat detection.

Technical Feasibility

- Open-source technology stack
- React + FastAPI architecture
- PyTorch Transformer + Isolation Forest
- REST-based modular services
- CPU-ready with optional GPU acceleration
- Cloud-deployable architecture

Potential Challenges

- Behaviour patterns evolve over time (concept drift)
- Limited labelled insider-threat datasets
- False positives affecting analyst trust
- Scaling to enterprise telemetry volumes
- Balancing detection accuracy with low inference latency

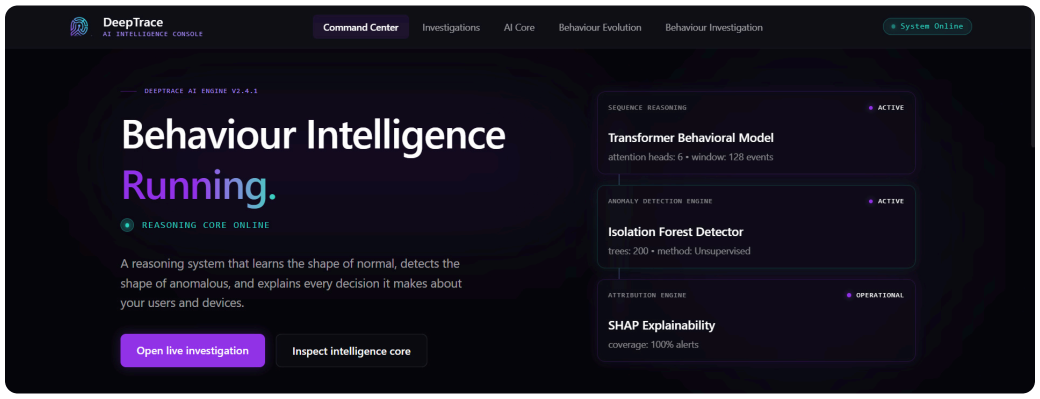
Risk Mitigation

- Behaviour Evolution adapts to changing patterns
- Hybrid AI combines Transformer and Isolation Forest
- SHAP explanations improve analyst confidence
- Modular architecture enables horizontal scaling
- Continuous model updates using new behavioural telemetry

IMPLEMENTATION

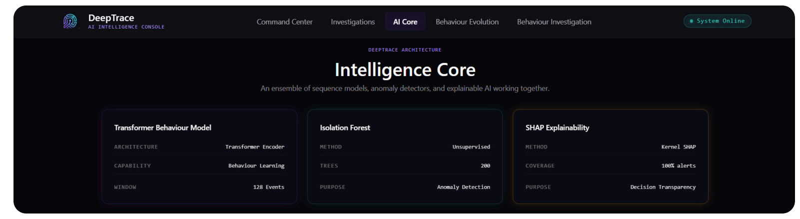
Implementation Artifacts

DeepTrace showcases a robust, full-stack implementation, demonstrating both front-end visualization and back-end AI logic.



Landing Page

Modern enterprise entry point with live project overview and navigation.



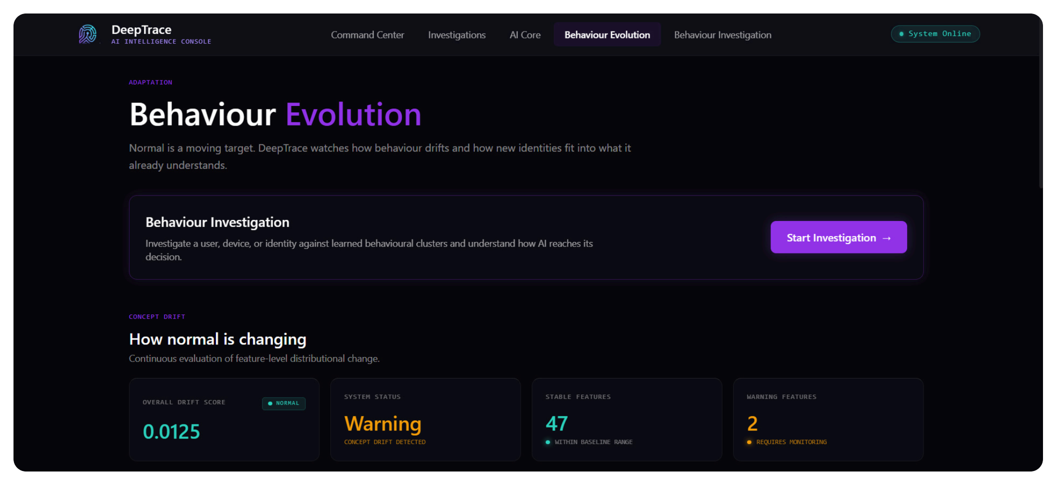
AI Core Dashboard

Real-time behavioural threat analysis powered by Transformer and Isolation Forest models.

```python
def predict_threat_risk(features: list) -> float:
    # 1. Feature Preprocessing
    X_scaled = scaler.transform(np.array([features]))

    # 2. Anomaly Detection (Isolation Forest)
    anomaly_score = iso_forest.decision_function(X_scaled)[0]

    # 3. Deep Learning Behaviour Encoder (Transformer)
    X_tensor = torch.tensor(X_scaled, dtype=torch.float32).unsqueeze(1)
    transformer_prob = transformer.predict_proba(X_tensor)

    # 4. Threat Fusion Engine
    # Normalize anomaly score (0-1) where lower is more anomalous
    normalized_anomaly = max(0.0, min(1.0, 0.5 - anomaly_score))

    # Combine scores using ensemble weighting
    fusion_score = (normalized_anomaly * 0.4) + (transformer_prob * 0.6)

    return round(fusion_score * 100, 2)
```

Core AI Inference Implementation

Hybrid AI pipeline combining feature preprocessing, anomaly detection, Transformer inference, and Threat Fusion.



Behaviour Evolution Monitoring

Continuous behavioural adaptation for detecting evolving insider threat patterns.

Artifacts demonstrating the implemented AI engine, enterprise dashboard, behaviour analytics, and explainable threat detection workflow.

IMPLEMENTATION

Research & References

The following references supported the research, implementation, and deployment of DeepTrace.

Research References

1. **Attention Is All You Need** (Transformer Architecture)
   https://arxiv.org/abs/1706.03762
2. **SHAP Documentation**
   https://shap.readthedocs.io/
3. **Isolation Forest** (Scikit-learn)
   https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html
4. **MITRE ATT&CK Framework**
   https://attack.mitre.org/
5. **PyTorch Documentation**
   https://pytorch.org/docs/stable/
6. **FastAPI Documentation**
   https://fastapi.tiangolo.com/

Project Resources

- **GitHub Repository**
  https://github.com/somiya-namdeo/DeepTrace
- **Frontend Live**
  https://deep-trace-steel.vercel.app/
- **Backend API**
  https://deeptrace-na69.onrender.com
- **Author**
  https://github.com/somiya-namdeo

DeepTrace is an AI-powered insider threat detection platform leveraging Transformer-based behaviour modelling, Isolation Forest anomaly detection, and explainable AI for enterprise behavioural security.